Galaxy Morpho

Team 7

***Abstract*—This project investigates galaxy morphology classification through a combination of theoretical study, computational modeling, and simulation. We begin with an exploration of the physical processes governing galaxy formation and evolution, supported by numerical calculations of key dynamical quantities. Building on this foundation, we develop a machine learning pipeline designed for large unlabeled astronomical datasets. The approach involves splitting the data by sky region, performing detailed feature engineering on both scalar and structured measurements, learning compact embeddings through unsupervised neural networks, applying clustering to generate pseudo-labels, and refining predictions with supervised classifiers trained on high-confidence samples. Finally, we perform an N-body gravitational simulation to examine how varying initial angular momentum influences the emergence of different galactic morphologies. This integrated framework provides both physical insight and robust classification tools applicable to future large-scale surveys**

Introduction

Galaxies are the fundamental building blocks of the observable universe—vast systems composed of stars, gas, dust, and dark matter. They exhibit a rich variety of forms, from the flattened disks of spiral galaxies to the spheroidal structures of ellipticals. Understanding how galaxies attain their morphology is a central question in astrophysics, one that connects processes from quantum fluctuations in the early universe to star formation, feedback, and gravitational interactions.

The study of galaxy morphology not only reveals the history of individual galaxies but also serves as a probe of cosmological structure formation. The processes behind morphological diversity involve dark matter halos, gas accretion, mergers, and environmental effects, and require both theoretical modeling and data-driven approaches to quantify.

In Module 1, we outline the theoretical foundation for galaxy formation and evolution, including the influence of dark matter, gravitational instability, angular momentum, and environment. This is complemented by numerical calculations of key dynamical quantities such as circular velocity, orbital period, and free-fall time.

Module 2 focuses on classifying galaxies in large unlabeled survey datasets. We split the data into sky regions to account for observational biases, engineer features from both scalar and list-type measurements, and apply standardization. Unsupervised neural network embeddings are learned, clustered to generate pseudo-labels, and refined by training supervised classifiers on high-confidence samples. Domain-specific parameters like Sersic index and axis ratio link clusters to physical galaxy types.

In Module 3, an N-body simulation demonstrates how varying initial angular momentum influences galaxy structure—high angular momentum forming disk-like spirals, and low angular momentum yielding spheroidal ellipticals.

By integrating cosmological physics with modern machine learning, this project offers a comprehensive exploration of how galaxies form and how we study them. It serves as both a theoretical investigation and a practical exercise in data-driven astrophysical discovery.

Module 1 : Formation of Galaxies

Task 1: Comprehensive description on formation of galaxies from the early universe to present day

1. Dark Matter Framework:

The Universe began in a nearly homogeneous and isotropic state shortly after the Big Bang. However, during the inflationary epoch, an extremely brief period of rapid exponential expansion, quantum fluctuations in the inflaton field were stretched to cosmological scales. These fluctuations introduced tiny inhomogeneities in the matter density field.

After inflation ended, these inhomogeneities were 'frozen in' as perturbations in the metric and the content of matter radiation. They are directly observed today as anisotropies in the cosmic microwave background (CMB), a snapshot of the Universe $\sim$380,000 years after the Big Bang. These fluctuations serve as the initial seeds for all later cosmic structure, from galaxies and clusters to the grand-scale cosmic web.

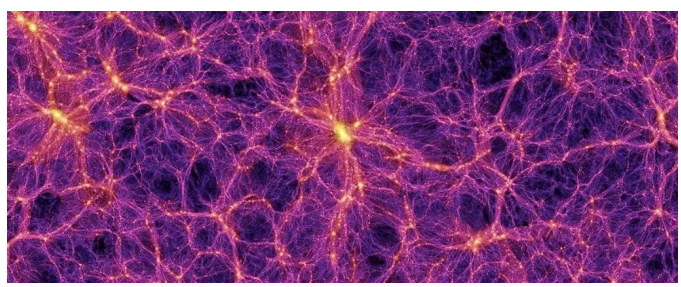

Fig. 1: Cosmic Structure

Linear Growth of Density Fluctuations

Once the Universe entered the matter-dominated era, gravitational instability began to amplify these primordial fluctuations. The evolution of the density contrast, defined as

$$\delta(\vec{x}, t) = \frac{\rho(\vec{x}, t) - \bar{\rho}(t)}{\bar{\rho}(t)},$$

is governed (for pressureless matter) by the linearized perturbation equation:

$$\ddot{\delta} + 2H\dot{\delta} = 4\pi G\bar{\rho}\delta,$$

where $H = \dot{a}/a$ is the Hubble parameter and $\bar{\rho}(t)$ is the background density.

In an Einstein–de Sitter Universe ($\Omega_m = 1$), the scale factor evolves as $a(t) \propto t^{2/3}$, and the growing mode of the solution becomes:

$$\delta(t) \propto a(t).$$

In this linear regime, the overdensities grow slowly but steadily. The overdense regions become denser by attracting more matter, while the underdense regions expand more quickly, forming voids. Although initially small, these perturbations gradually reach the point where a nonlinear gravitational collapse begins to dominate.

This stage sets up a predictable and orderly growth of structure, a blueprint that tells us which mass scales will collapse first and when. The statistical properties of this growth are encoded in the matter power spectrum $P(k)$, which determines the relative strength of fluctuations at spatial scales.

Nonlinear Collapse and the Role of the Virial Theorem

As the Universe evolves, some regions of space develop large density contrasts ($\delta \gtrsim 1$) where linear theory fails. These regions begin to decouple from cosmic expansion, and gravitational collapse drives them toward forming bound structures.

To understand this, consider the spherical top-hat collapse model, a simplified but powerful approach in which a uniform spherical overdensity evolves under its own gravity. The equation of motion is Newtonian:

$$\frac{d^2R}{dt^2} = -\frac{GM}{R^2},$$

where $R(t)$ is the radius of the sphere and M its mass. Initially, the overdensity expands with the Hubble flow, but eventually slows, reaches a maximum radius $R_{\rm ta}$ at turnaround, and then collapses.

At the end of collapse, however, the region does not form a singularity. Instead, gravitational interactions cause energy redistribution (violent relaxation), and the system reaches a virial equilibrium, described by the Virial Theorem:

$$2K + U = 0,$$

where K is kinetic energy and U is gravitational potential energy.

Conservation of total energy from turnaround to virialization gives:

$$U_{\rm vir} = 2U_{\rm ta} \Rightarrow R_{\rm vir} = \frac{1}{2}R_{\rm ta}.$$

The resulting virialized object — a dark matter halo — has a stable size and internal structure. The average overdensity within this radius is known as the virial overdensity, and for an Einstein–de Sitter universe is approximately:

$$\Delta_{\rm vir} = \frac{\rho_{\rm vir}}{\bar{\rho}} \approx 18\pi^2 \approx 178.$$

This sets the physical criteria used in simulations to identify bound structures.

Dark Matter Halos: Properties and Internal Structure

The dark matter halo is a gravitationally bound, collisionless system that forms from nonlinear collapse. It is the foundational unit of cosmic structure, serving as the environment where galaxies form and evolve.

N-body simulations reveal that dark matter halos exhibit a universal internal density profile — the Navarro–Frenk–White (NFW) profile:

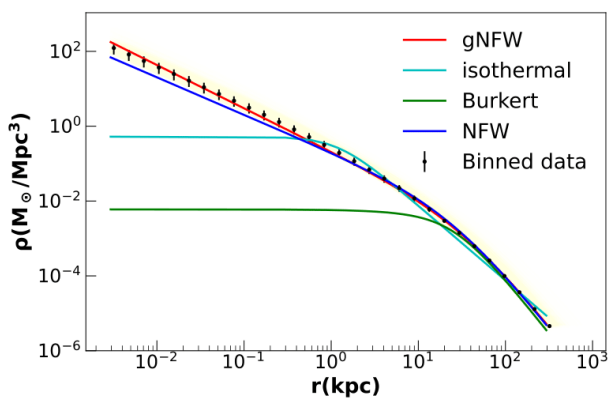

Fig. 2: An example of Dark Matter Density Profiles

$$\rho(r) = \frac{\rho_s}{(r/r_s)(1 + r/r_s)^2},$$

where ρ_s and r_s are the scale density and radius, respectively.

This profile has two asymptotic behaviors:

$$\rho(r) \propto r^{-1} \quad \text{for small } r \quad \text{(cuspy core)},$$
$$\rho(r) \propto r^{-3} \quad \text{for large } r \quad \text{(rapid decline)}.$$

The concentration parameter, defined as $c = R_{\rm vir}/r_s$, indicates how dense the halo core is relative to its size, and correlates with its formation time: earlier-forming halos are more concentrated.

Halos provide deep potential wells into which baryonic gas can fall, shock-heat to the virial temperature, cool, and form stars. Thus, halos are not only gravitational products of early fluctuations — they are the sites where visible galaxies are born.

Statistical Abundance of Halos: Press–Schechter Formalism

To predict the number of halos of a given mass, we turn to the Press–Schechter formalism — a statistical framework grounded in the early Gaussian random field of density perturbations.

The approach is based on the idea that regions whose smoothed overdensity exceeds a critical value $\delta_c \approx 1.686$

(from spherical collapse) will collapse to form halos. The halo mass function, or number density of halos per unit mass, is:

$$\frac{dn}{dM} = \frac{2\bar{\rho}}{M^2} \frac{\delta_c}{\sigma(M)} \left| \frac{d\ln\sigma}{d\ln M} \right| \exp\left(-\frac{\delta_c^2}{2\sigma^2(M)} \right),$$

where:

- $\sigma(M)$: RMS density fluctuation on mass scale M,
- $\bar{\rho}$: mean matter density.

This function predicts that low-mass halos form earlier and are more numerous, while high-mass halos form later and are rare.

This theoretical mass function has been validated by large-scale simulations and is essential for connecting the initial conditions of the Universe to the observed distribution of galaxies and clusters.

Dark matter determines where and when galaxies form, sets the stage for baryonic processes, and governs the mass and angular momentum budget that ultimately shape galaxy morphology.

[1][2][3][4][5]

2. Baryonic Physics:

In the early universe, matter existed in a nearly uniform distribution, with only slight density fluctuations imprinted during the inflationary epoch. These fluctuations served as seeds for gravitational instability. As dark matter dominated the mass budget of the universe, it collapsed first under gravity, forming potential wells known as dark matter halos. Baryonic matter—composed of atoms, ions, and electrons—was then gravitationally attracted into these halos, where it underwent a complex series of physical processes that led to galaxy formation.

Baryons Falling into Dark Matter Halos

After recombination, the universe became transparent and neutral, and baryons were able to fall freely under gravity. These baryons, unlike collisionless dark matter, are collisional and interact via pressure and radiation. As they entered the gravitational wells formed by dark matter, baryonic gas was compressed and shock-heated to the virial temperature of the halo, often reaching values from 10^4 to 10^7 K, depending on the halo mass:

$$T_{\text{vir}} \approx \frac{\mu m_p V_c^2}{2k_B} \sim 10^6\,\text{K} \left(\frac{V_c}{167\,\text{km/s}} \right)^2$$

Here:

- μ is the mean molecular weight,
- m_p is the proton mass,
- k_B is Boltzmann's constant,
- V_c is the circular velocity of the halo.

This process created an extended hot gaseous halo in quasi-hydrostatic equilibrium with the surrounding dark matter.

Radiative Cooling and Cloud Formation

For stars to eventually form, this hot gas must cool and condense toward the center of the halo. Cooling proceeds via radiative processes:

(a) Atomic Line Cooling:

At temperatures between 10^4–10^6 K, cooling is dominated by collisional excitation and ionization of hydrogen and helium.

Photons are emitted when excited electrons transition to lower energy states.

This mechanism becomes inefficient below 10^4 K in primordial gas.

(b) Metal Line Cooling:

In gas enriched by previous stellar generations (Pop III supernovae), heavier elements (C, O, Fe) provide additional line transitions.

Metal-line cooling significantly enhances the cooling rate at $T \lesssim 10^5$ K, enabling more efficient gas condensation.

(c) Molecular Cooling:

At low temperatures ($T \lesssim 10^3$ K), the dominant coolant is molecular hydrogen (H_2).

H_2 forms through residual free electrons acting as catalysts in reactions like:

$$\text{H} + e^- \rightarrow \text{H}^- + \gamma, \quad \text{H}^- + \text{H} \rightarrow \text{H}_2 + e^-$$

H_2 enables cooling to temperatures of a few hundred Kelvin, essential for fragmentation and gravitational collapse.

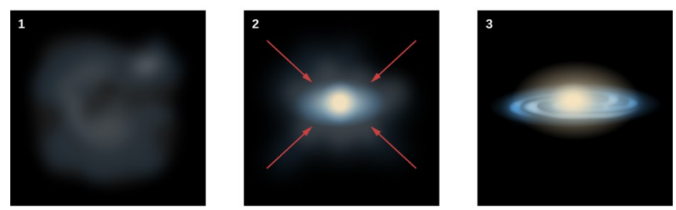

Fig. 3: Galaxy Formation

This radiative cooling reduces gas pressure, allowing it to sink deeper into the halo potential well, eventually forming a rotating disk or central bulge.

Formation of Molecular Clouds and Star Formation

As gas condenses and cools further, it forms dense, self-gravitating regions within the halo known as molecular clouds. These are the primary birthplaces of stars.

Once local density is sufficient, gravity overwhelms internal pressure and turbulence, triggering Jeans instability:

$$M_J \sim \left(\frac{5k_B T}{G\mu m_p} \right)^{3/2} \left(\frac{3}{4\pi\rho} \right)^{1/2}$$

This gives the minimum mass scale at which a gas cloud will collapse under its own gravity.

Fragmentation of these molecular clouds leads to protostellar cores. As they collapse, potential energy is converted to heat, raising central temperatures.

When core temperatures exceed $\sim 10^6$ K, hydrogen fusion ignites, marking the birth of a star.

Starbursts in the Early Universe

In the early universe, especially within the first gigayear:

- Pristine gas (mostly hydrogen and helium) was abundantly available,
- Halos were smaller and more abundant (from hierarchical formation),
- Cooling times were short compared to dynamical times.

These conditions led to bursty star formation phases — the rapid formation of stars in compact regions, laying the groundwork for the first galaxies and chemical enrichment of the interstellar medium (ISM).

These processes also began the feedback cycle, where stellar winds and supernovae returned energy and metals to the surrounding gas, regulating subsequent star formation — a mechanism discussed in later section.

3. Stellar Evolution:

The evolution of stars in the early universe is a cornerstone of cosmic history. The first stars not only illuminated the darkness of the post-recombination universe but also initiated a chain reaction that fundamentally shaped the chemical composition and structure of galaxies. This transition — from the first Population III (Pop III) stars to Population II (Pop II) and later Population I stars — is critical for understanding the metal enrichment of the interstellar medium (ISM) and the progression of galactic evolution.

Formation of Population III Stars: Stars of the Pristine Universe

In the early universe, gas consisted almost entirely of hydrogen and helium, formed during Big Bang nucleosynthesis. The absence of heavier elements or metals in astronomical terms had profound implications for the formation of the first stars.

(a) Cooling Challenge in Metal-Free Gas:

Without metals or dust to aid cooling, gas clouds in dark matter halos could only cool through molecular hydrogen (H_2) rotational and vibrational transitions, which are inefficient at low densities and temperatures. This limited the ability of gas to fragment into small clumps, leading to the formation of very massive stars — typically 10 to 300 $M_\odot$.

(b) Physical Characteristics:

Pop III stars were:

- Extremely massive
- Extremely hot and luminous
- Short-lived (a few million years)
- Formed in mini-halos of mass $\sim 10^5$–$10^6\ M_\odot$ where H_2 cooling could occur

(c) Stellar Evolution and Fate:

- Below 140 $M_\odot$: Core-collapse supernovae
- 140–260 $M_\odot$: Pair-instability supernovae, leaving no remnant and dispersing all metals
- Above 260 $M_\odot$: Direct collapse to black holes

The significance of Pop III stars lies in their nucleosynthesis: these stars forged the first heavy elements (C, O, Fe...) via fusion and released them through supernovae, enriching their surrounding medium — a process that fundamentally transformed the chemistry of the cosmos.

[3][4][5]

Transition to Population II Stars:

As Pop III stars exploded, they seeded their environments with metals. This led to a chemical transformation of the ISM and the birth of Population II stars.

(a) Metallicity Threshold for Transition:

Once metallicity reached a critical value $Z_{\text{crit}} \sim 10^{-4} - 10^{-3}\ Z_\odot$, gas clouds could cool more efficiently via metal-line and dust cooling, leading to more fragmentation and lower-mass star formation.

(b) Properties of Pop II Stars:

- Low metallicity (but not zero)
- Longer lifetimes than Pop III (several Gyr)
- Found in galactic halos and globular clusters
- Played a key role in the reionization epoch and in continuing metal enrichment

As metals accumulated, the process of star formation became more efficient and less top-heavy. The average stellar mass decreased over time, and star formation shifted from rare massive bursts to more continuous, sustained processes.

Population I Stars and the Modern ISM

Eventually, as more stellar generations lived and died, the ISM became heavily enriched. Population I stars, including the Sun, formed from this metal-rich gas.

Pop I stars have:

- Metallicities close to solar (Z $\sim Z_\odot$)
- Greater potential for forming planetary systems
- Lifetimes and masses spanning wide ranges
- Locations primarily in galactic disks

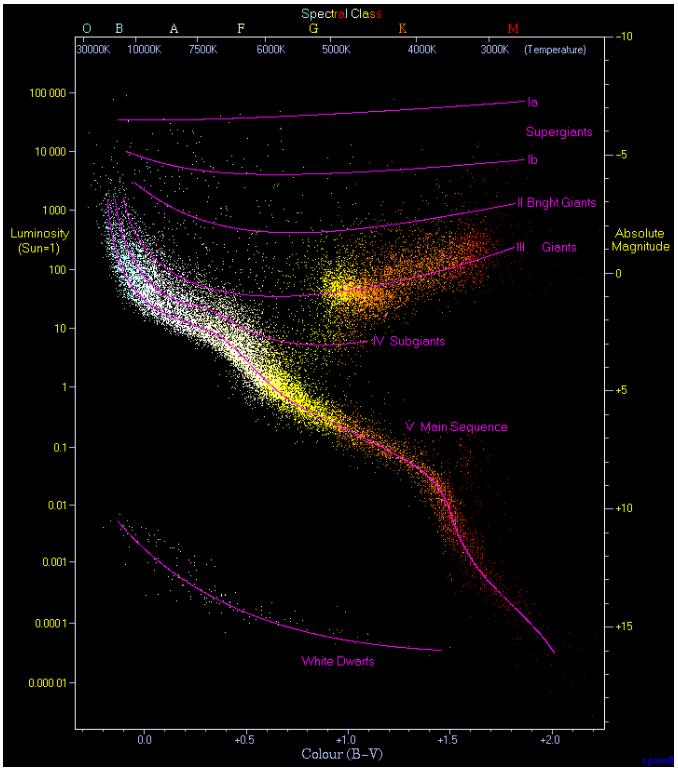

Fig. 4: Hertzsprung-Russell diagram

This progressive increase in metallicity is referred to as chemical evolution, and it dramatically shaped star formation, galaxy morphology, and the emergence of complex structures such as protoplanetary disks and life-sustaining planets.

Impact of Stellar Evolution on Galaxy Evolution

The evolution from Pop III $\rightarrow$ Pop II $\rightarrow$ Pop I created a feedback loop that drove galactic transformation:

TABLE I: Stellar Generations and Their Properties

Generation	Lifetime	Enrichment	Legacy
Pop III	Few Myr	First heavy elements	Supernovae, BH seeds
Pop II	Gyr	Enrich ISM	Globular clusters, halo stars
Pop I	Gyr+	Ongoing	Planetary systems, life

ISM Metallicity & Cooling:

Metals introduced by stellar evolution enhanced cooling rates, shifting the IMF (Initial Mass Function) toward lower masses and altering the star formation efficiency.

Supernova Explosions:

Returned enriched gas to the ISM, triggered new star formation, and contributed to feedback processes (next section).

Galactic Morphology:

Starburst activity in early phases shaped bulge-disk structures, and chemical gradients influenced galaxy color and spectrum.

[6][3][4][5]

4. Feedback Processes:

Star formation in galaxies is not a one-way process. Left unchecked, gas cooling and gravitational collapse would produce stars at rates far exceeding what we observe. Instead, feedback mechanisms—ranging from stellar winds to cataclysmic supernovae and the violent outbursts from active galactic nuclei (AGN)—inject energy, momentum, and metals back into the interstellar and circumgalactic media. These feedback processes regulate star formation, prevent runaway collapse, and shape the structural and chemical evolution of galaxies over cosmic time.

Stellar Feedback: Winds, Radiation, and Supernovae

(a) Young Stars and UV Radiation

The first regulatory force comes from massive young stars. These stars emit intense ultraviolet (UV) radiation that ionizes surrounding gas, creating H II regions. This photoionization heats the gas to $\sim 10^4$ K, raising pressure and driving photoevaporation flows that clear the gas near the stars.

Moreover, massive stars lose mass through stellar winds—high-speed outflows of plasma accelerated by radiation pressure. These winds carry momentum and energy into the ISM, stirring turbulence and driving shock fronts.

UV radiation and winds can:

- Evacuate cavities in star-forming clouds.
- Disrupt molecular cloud cores, halting further collapse.
- Trigger sequential star formation in neighboring regions via compression.

(b) Supernova Explosions

The most dramatic form of stellar feedback comes from supernovae (SNe), particularly those from massive stars (Type II) and white dwarf accretion (Type Ia). A single supernova can release $\sim 10^{51}$ ergs of energy—enough to heat and shock the surrounding ISM significantly.

Shock-heated gas expands and pushes out the surrounding medium, driving galactic winds in low-mass galaxies where gravitational binding is weak.

Supernova-driven turbulence regulates cloud fragmentation, effectively influencing the Initial Mass Function (IMF).

The chemical yields of supernovae enrich the ISM with metals, fueling later stellar generations with more efficient cooling pathways.

Supernova feedback can be either:

- Positive (triggering star formation in dense shocked shells), or
- Negative (disrupting clouds and halting star formation).

This dual role is key to the self-regulating nature of galaxies. [7]

AGN Feedback: Supermassive Black Holes as Cosmic Regulators

While stellar feedback dominates in small and medium galaxies, the most massive systems require a much more powerful regulator: active galactic nuclei (AGN) powered by accreting supermassive black holes (SMBHs).

(a) Accretion and Energy Release

When cold gas accretes onto an SMBH, a portion of its gravitational energy is converted into radiation and kinetic energy. This process results in:

- Broadband radiation (X-rays, UV, optical)
- Relativistic jets
- Hot ionized outflows and radio lobes

Although SMBHs are tiny compared to galaxies, their energy output can rival or exceed the binding energy of the host galaxy's gas.

(b) Feedback Modes

AGN feedback operates primarily in two modes:

Radiative (quasar) mode:

- Occurs at high accretion rates.
- Strong radiation pressure heats and ionizes the gas, quenching star formation.
- Often associated with luminous quasars at high redshift.

Kinetic (radio) mode:

- Dominant in massive ellipticals and clusters.
- Relativistic jets inflate radio bubbles in the hot halo gas.
- Prevents cooling flows from fueling more star formation.

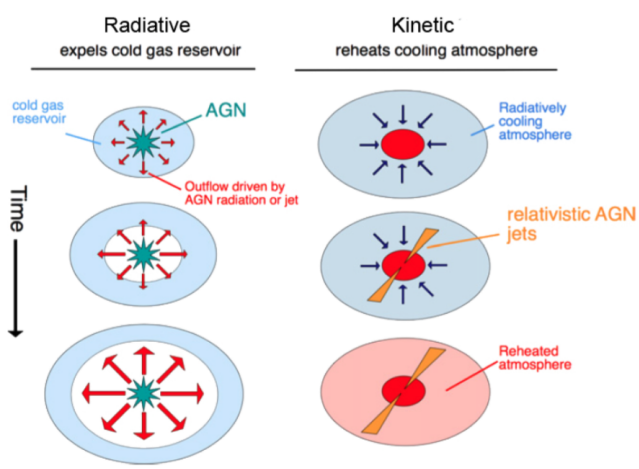

Fig. 5: Two modes of AGN feedback

(c) Quenching Star Formation

AGN-driven outflows can:

- Heat the circumgalactic medium, preventing fresh gas infall.
- Expel cold gas reservoirs, directly halting new star formation.
- Suppress cooling in hot halos (particularly in galaxy clusters).

Simulations including AGN feedback match observations of:

- The galaxy luminosity function cutoff.
- The color bimodality (red vs. blue galaxies).
- The mass-metallicity relation.

[8]

Regulation and Self-Balancing of Galaxy Growth

Together, stellar and AGN feedback enforce a delicate balance between gas accretion, cooling, and star formation:

Process	Effect
Gas accretion	Fuels star formation
Star formation	Generates radiation, winds, and SNe
Feedback	Regulates further gas cooling and inflow

This feedback loop is crucial. Without it:

- Simulations overpredict star formation and galaxy masses.
- Cooling catastrophes occur in massive halos.

With it:

- Galaxy stellar masses align with halo mass.
- Star formation is quenched in massive systems.
- Observed morphologies and colors are reproduced.

Mathematically, the feedback effect can be captured in hydrodynamic simulations via:

$$\frac{dE_{\text{feedback}}}{dt} = \epsilon \cdot \dot{M}_* \cdot c^2$$

where ϵ is the feedback efficiency, $\dot{M}_*$ is the star formation rate, and c is the speed of light (AGN case). This term adds pressure, turbulence, and heating in simulation cells, mimicking real astrophysical feedback.

[3][4][5]

5. Hierarchical Assembly:

Introduction: The Bottom-Up Nature of Cosmic Structure Formation

In the standard ΛCDM cosmological model, structure formation proceeds hierarchically — small structures form first and progressively merge or accrete mass to create larger systems. This bottom-up evolution stems from the scale-invariant nature of primordial fluctuations generated during cosmic inflation. These initial overdensities collapse to form the first small dark matter halos, which serve as gravitational wells for baryonic matter. Over time, these halos merge — through both major (comparable mass) and minor (massive halo absorbing a smaller one) mergers — creating progressively larger systems that ultimately host galaxies, groups, and clusters.

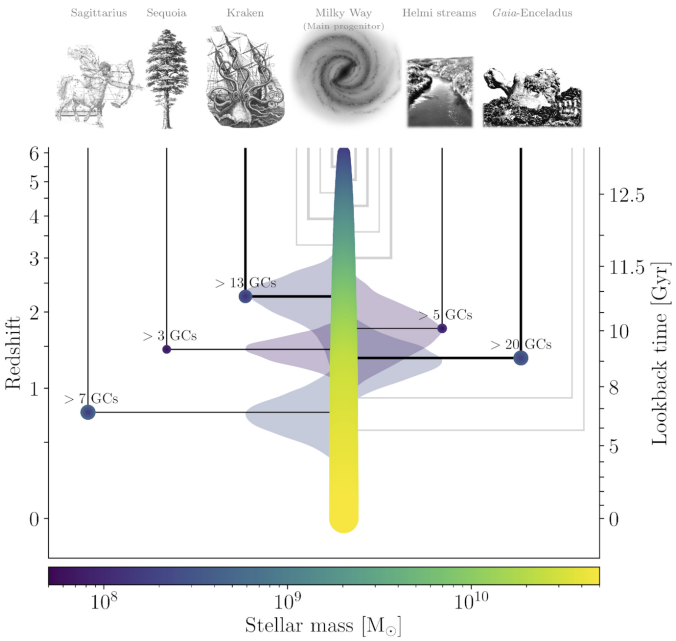

Fig. 6: Galaxy merger tree of the Milky Way

This hierarchical buildup is the fundamental mechanism by which the complex cosmic web and diverse galaxy morphologies arise. Unlike the top-down scenario (now ruled out), in which large structures fragmented into smaller units, the bottom-up picture aligns with both simulations and large-scale galaxy surveys.

The Dark Matter halos, once formed, do not remain isolated — they interact, merge, or accrete additional matter. These processes are described using merger trees, where smaller progenitor halos branch out and join larger descendants. Numerical simulations such as the Millennium Simulation vividly demonstrate this process, showing how large-scale structures like the Sloan Great Wall arise from billions of years of gravitational evolution.

Major and Minor Mergers: Core of Hierarchical Assembly

Mergers between dark matter halos — and their baryonic galaxies — are the cornerstone of hierarchical growth:

(a) Major mergers

(mass ratios ∼1:1 to 1:3) lead to violent dynamical interactions. The disks of spiral galaxies are typically destroyed, and the system undergoes a burst of star formation, followed by violent relaxation. The result is often a spheroidal galaxy (elliptical or lenticular). The randomization of stellar orbits in such events explains the lack of organized rotation in these galaxies.

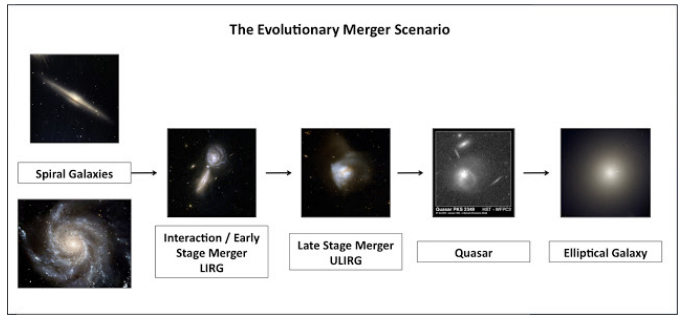

Fig. 7: Major merger scenario

(b) minor mergers

(mass ratios ≥1:10) are more frequent and usually preserve the disk structure of the primary galaxy. However, they can lead to the gradual buildup of spheroidal components (bulges), heating of the stellar disk, and accretion of fresh gas and satellites.

Over cosmic time, these repeated mergers not only increase galaxy mass but also drive morphological transformation. This gives rise to the Hubble sequence, where early-type galaxies (ellipticals) dominate high-density regions with high merger rates, and late-type galaxies (spirals) dominate the field.

Starbursts, AGN, and Feedback During Mergers

Mergers trigger gravitational torques that funnel gas towards the galactic center. This inflow can lead to:

- Starbursts: rapid, short-lived episodes of star formation, observed in systems like ULIRGs.
- AGN activity: as gas accretes onto the central supermassive black hole, it produces energetic feedback that can heat or eject the remaining gas, quenching future star formation.

This interplay — between star formation and AGN — is vital in regulating the mass growth and morphology of galaxies during mergers.

Satellite Accretion, Tidal Disruption, and Halo Substructure

Hierarchical assembly also includes continuous accretion of satellite galaxies, often observed as:

- Stellar streams in the halos of Milky Way and Andromeda.
- Tidal tails and shells in merger remnants.
- Heating and thickening of galactic disks by repeated satellite infall.

The subhalo population preserved in N-body simulations contributes to this growth and to the rich substructure observed in galaxies today. While some satellites are tidally disrupted and contribute to stellar halos, others survive as dwarf galaxies.

This explains the multi-component nature of galaxies: disks, bulges, stellar halos, and globular clusters are all partly shaped by this process.

Environmental Dependence and Morphological Diversity

The hierarchical framework naturally explains environmental effects:

In dense environments (clusters), high merger rates lead to elliptical-dominated populations.

In low-density environments, spiral galaxies persist, having undergone fewer major mergers.

It also explains the morphology-density relation, where galaxy morphology correlates with local density — a direct consequence of merger-driven evolution.

Simulation Insights and Observational Evidence

Large simulations (e.g., Millennium, Illustris, EAGLE) demonstrate how merging halos evolve into realistic galaxy populations. These match observed luminosity functions, color–magnitude diagrams, and morphology distributions when feedback processes are included.

Observational support comes from:

- Deep surveys (CANDELS, COSMOS) showing high merger rates at $z > 1$,
- Kinematic studies showing rotating disks transitioning into dispersion-dominated systems,
- Morphological statistics from HST and SDSS.

[9][3][4][5]

6. Environmental Influences:

The evolution of galaxies is not determined solely by their initial conditions and internal processes; it is also strongly influenced by their cosmic environment. Galaxies can reside in vastly different surroundings — from isolated low-density regions (the field) to dense gravitationally bound structures containing hundreds or thousands of members (galaxy clusters). These environments subject galaxies to distinct gravitational, hydrodynamic, and radiative interactions, altering their morphology, gas content, and star formation histories.

Cluster Environments:

(a) Properties of Clusters

- **Mass:** $\sim 10^{14} - 10^{15}\, M_\odot$
- **Population:** Hundreds to thousands of galaxies, bound by the cluster's dark matter halo.
- **Intracluster Medium (ICM):** Filled with hot, X-ray emitting plasma at $T \sim 10^7 - 10^8$ K, with densities of $10^{-3} - 10^{-4}\,\mathrm{cm}^{-3}$.
- **Dynamics:** Galaxies move at velocities of 500–1500 km/s relative to the cluster center.

The cluster potential well and dense ICM create several environmental quenching mechanisms:

(b) Ram-Pressure Stripping

As galaxies move through the ICM, the hot gas exerts a ram pressure:

$$P_{\mathrm{ram}} = \rho_{\mathrm{ICM}} v_{\mathrm{gal}}^2$$

If P_{ram} exceeds the gravitational restoring force on the gas disk of the galaxy, the cold gas is swept out, stopping star formation. This process explains why many cluster spirals are "anaemic" — retaining their stellar disks but devoid of star-forming gas.

Fig. 8: Tails in spiral galaxy D100 created by ram-pressure stripping

(c) Tidal Interactions and Galaxy Harassment

Clusters have high galaxy densities, leading to:

- **Tidal stripping:** Removal of outer stars and dark matter halos during close encounters with the cluster potential or other galaxies.
- **Galaxy harassment:** Cumulative effect of many high-speed flybys, distorting disks, triggering central starbursts, or even transforming spirals into spheroidals.

(d) Strangulation (Starvation)

Infalling galaxies lose their extended hot gas halo to the ICM, cutting off future gas accretion. Star formation fades once the remaining cold gas is exhausted.

(e) Morphological Outcome

- High fraction of elliptical and S0 galaxies ("red and dead")
- Low star formation rates, older stellar populations
- Smooth, pressure-supported morphologies

Field Galaxies:

(a) Properties

- **Environment:** Low galaxy density; not bound to massive cluster potentials.
- **Dynamics:** Lower relative velocities between neighbors ($\sim$ 100–300 km/s).
- **Gas Reservoirs:** Retain both cold and hot gas halos, allowing prolonged accretion.

(b) Evolutionary Pathways

- More gradual evolution, with mergers typically being major but infrequent.
- Ongoing gas accretion from the intergalactic medium sustains continuous star formation.
- Preservation of ordered rotational motion supports disk structures.

(c) Morphological Outcome

- Higher proportion of spiral and irregular galaxies
- Blue colors from young stellar populations
- Greater diversity in morphology due to stochastic merger history

Morphology–Density Relation

One of the most robust observational results in extragalactic astronomy is the morphology–density relation:

- High-density regions (clusters): Ellipticals and S0 galaxies dominate.
- Low-density regions (field): Spirals and irregulars dominate.

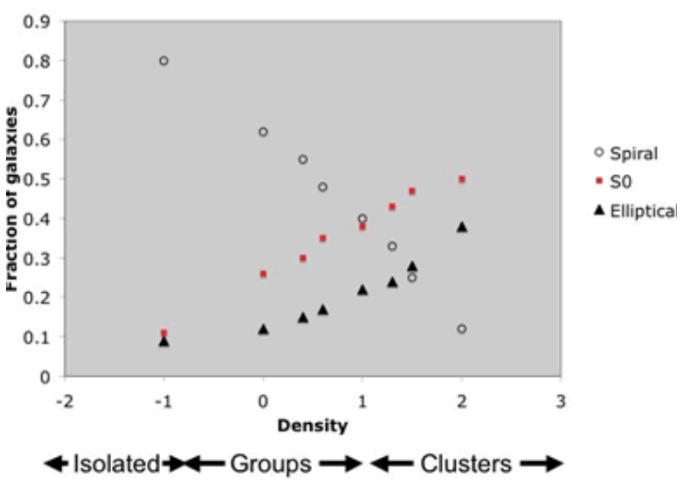

Fig. 9: Morphology Density relation

This relation reflects the combined action of:

- Internal processes: Secular evolution, major mergers
- External processes: Ram-pressure stripping, tidal forces, starvation

Its persistence across redshift suggests that environmental effects operate over cosmic time and contribute significantly to galaxy diversity.

Environmental Influence in the Cosmic Web Context

Clusters are located at the nodes of the cosmic web, where filaments intersect, funneling both galaxies and gas inward. The infall region around clusters is a site of pre-processing:

- Groups of galaxies enter clusters already altered by group-scale tidal and gas interactions.
- Field galaxies, in contrast, evolve within filaments and sheets, influenced more by cold-gas accretion and isolated mergers.

[10][3][4][5]

Task 2: Using Newtonian physics to demonstrate why galaxies tend to flatten into disk structures

Part A - Angular Momentum Conservation

A fluid element in a rotating cloud has angular momentum about the rotation axis. If the angular momentum is conserved during collapse, the tangential velocity must increase as the radius shrinks. The resulting centrifugal force grows faster with decreasing radius than gravity does, so the centrifugal force becomes dominant and halts further radial collapse. In the end the material collapses into a flattened, rotating structure (a disk).

Set-up and notation

Consider an initially spherical gas cloud of total mass M and initial radius R_i.

Choose cylindrical coordinates (R, ϕ, z) with rotation about the z-axis.

A fluid parcel initially at radius R has velocity v with tangential component v_ϕ.

Specific angular momentum of that parcel (about z-axis) is

$$j_z = (\mathbf{r} \times \mathbf{v}) \cdot \hat{\mathbf{z}} = R v_\phi.$$

We assume axis-symmetric rotation with no external torques acting on the parcel during the (local) collapse, so j_z is conserved for that parcel.

Conservation of specific angular momentum

If j_z is conserved during radial contraction from radius R to a smaller radius R',

$$j_z = R v_\phi(R) = R' v_\phi(R') = \text{constant}.$$

Thus the tangential velocity scales as

$$v_\phi(R') = \frac{j_z}{R'} \quad \Rightarrow \quad v_\phi \propto \frac{1}{R}.$$

Centrifugal vs gravitational acceleration

Centrifugal acceleration (per unit mass) for circular motion at radius R:

$$a_{\text{cent}} = \frac{v_\phi^2}{R}.$$

Using $v_\phi = j_z/R$,

$$a_{\text{cent}} = \frac{j_z^2}{R^3}.$$

So centrifugal acceleration grows as R^{-3} during contraction if j_z is conserved.

And we already know that.

$$a_{\text{grav}} \sim \frac{GM}{R^2}.$$

So gravity scales as R^{-2}. As the cloud contracts (decreasing R), a_{cent} increases faster ($\propto\ R^{-3}$) than a_{grav} ($\propto\ R^{-2}$). Therefore, even if centrifugal support is small initially, it becomes dominant at small radii and can halt further collapse in the radial (equatorial) direction.

Equating centripetal requirement to gravitational pull we find a radius R_c where centrifugal force can balance gravity:

$$\frac{j_z^2}{R_c^3} \sim \frac{GM}{R_c^2} \quad \Rightarrow \quad j_z^2 \sim GMR_c.$$

So the characteristic radius where rotation stalls inward collapse is

$$R_c \sim \frac{j_z^2}{GM}.$$

If the cloud contracts to $R \gg R_c$, gravity dominates and collapse continues.

When $R \lesssim R_c$, centrifugal support becomes comparable to gravity, the gas stops collapsing inward and instead starts to spin about the radius R_c.

This is the simple reason for more collapse in the equatorial plane than along the axis.

Along the rotation axis

Along the rotation axis (z-direction) the centrifugal term is zero because tangential motion does not provide support in z. A fluid element located off the mid-plane still feels a vertical gravitational component toward $z = 0$. Nothing analogous to a_{cent} prevents collapse along z, so collapse continues in the vertical direction to much smaller scales.

Therefore the cloud becomes flattened: dense in the equatorial plane (where centrifugal balance halts radial collapse) but thin in z (where collapse proceeds).

Vertical Thickness

Once rotation halts further radial collapse, vertical collapse toward the midplane continues until pressure (thermal or turbulent) balances the vertical component of gravity. For an isothermal gas with sound speed c_s in a rotating disk with angular frequency $\Omega(R) = v_\phi/R$, hydrostatic balance gives the characteristic scale height

$$H \sim \frac{c_s}{\Omega}$$

Thus, when rotation dominates ($v_\phi \gg c_s$), the structure is thin ($H \ll R$).

PART B - ENERGY CONSIDERATIONS

Quick definitions and notation

Total mass of the cloud: M.

Typical radius of interest: R.

Specific angular momentum (about the z-axis): $\mathbf{j} = \mathbf{r} \times \mathbf{v}$, scalar about z: $j = Rv_\phi$. We assume no external torque on the fluid parcels.

Total kinetic energy: T. Gravitational potential energy: W ($W < 0$). Moment of inertia about center: $I = \int r^2\, dm$.

Decompose kinetic energy per mass or globally into radial (infall), tangential (rotation), and random/thermal parts:

$$T = T_{\text{rad}} + T_{\text{rot}} + T_{\text{th}}.$$

The scalar virial theorem

The scalar virial theorem for a self-gravitating system in Newtonian gravity is obtained from the second time derivative of the moment of inertia I:

$$\frac{1}{2}\ddot{I} = 2T + W + S,$$

where S are surface terms (for an isolated cloud we typically ignore surface pressure terms or include them as needed). For a system in steady state (time-average or equilibrium), $\ddot{I} = 0$ and $S \approx 0$, so

$$2T + W = 0.$$

Thus in equilibrium the total kinetic energy equals half the magnitude of the gravitational potential energy:

$$T = -\frac{W}{2} = \frac{|W|}{2}.$$

During collapse, $\ddot{I} < 0$ and the virial relation shows gravitational binding energy change drives changes in T.

Differential view: how ΔW redistributes

Take an incremental collapse that changes the gravitational energy by ΔW. The differential form of the virial equation (ignoring surface terms) gives:

$$\frac{1}{2}\Delta\ddot{I} = 2\,\Delta T + \Delta W.$$

If the collapse happens slowly compared to the system's natural free-fall time, we can treat it as being almost in balance at each step (or averaged over small oscillations). In that case, the left-hand term is small on average, so we can write the relation approximately as

$$2\,\Delta T + \Delta W \approx 0 \quad \Rightarrow \quad \Delta T \approx -\frac{\Delta W}{2}.$$

When the system releases gravitational energy $|\Delta W|$ during collapse, roughly half of that energy (on average) goes into increasing kinetic energy T (bulk motions + rotation + random/thermal), while the other half must be radiated away or go into internal energy/heat if the system reaches a new equilibrium. This is the usual virial argument for why collapsing gas heats up.

Decomposing kinetic energy and scaling during collapse

Consider a parcel of mass m with conserved specific angular momentum j. When its radius contracts from R to a smaller R':

Tangential (rotational) kinetic energy of the parcel:

$$T_{\mathrm{rot,\,parcel}} = \frac{1}{2}mv_\phi^2 = \frac{1}{2}m\left(\frac{j}{R}\right)^2 = \frac{1}{2}m\frac{j^2}{R^2}.$$

So for a fixed j,

$$T_{\mathrm{rot}} \propto \frac{1}{R^2}.$$

It grows very rapidly as R shrinks.

Radial kinetic energy (rough estimate): if something falls inward from rest to a distance R, its speed is mostly determined by the pull of gravity at that distance.

$$v_{\mathrm{rad}} \sim \sqrt{\frac{2GM}{R}}$$
$$\Rightarrow \quad T_{\mathrm{rad,\,parcel}} \sim \frac{1}{2}mv_{\mathrm{rad}}^2 \sim \frac{1}{2}m\frac{2GM}{R} = m\frac{GM}{R}.$$

Thus,

$$T_{\mathrm{rad}} \propto \frac{1}{R}.$$

As the cloud collapses, tangential kinetic energy grows as R^{-2} while radial kinetic energy grows as R^{-1}. Therefore angular-momentum conservation channels a larger fraction of the released gravitational energy into increasing rotational kinetic energy at small radii. This is the core energetic reason why rotation halts radial collapse earlier than vertical (axial) collapse and contributes to disk formation.

Condition for rotational (centrifugal) support

A local force balance comparing centrifugal acceleration to gravitational acceleration yields the well-known condition for rotational support of a parcel at radius R:

Centrifugal acceleration:

$$a_{\mathrm{cent}} = \frac{v_\phi^2}{R} = \frac{j^2}{R^3}.$$

Gravitational acceleration (assuming roughly mass $M(< R)$ interior):

$$a_{\mathrm{grav}} \sim \frac{GM}{R^2}.$$

Requiring centrifugal support ($a_{\mathrm{cent}} \gtrsim a_{\mathrm{grav}}$) gives

$$\frac{j^2}{R^3} \gtrsim \frac{GM}{R^2} \quad \Rightarrow \quad j^2 \gtrsim GMR.$$

If this condition holds for a parcel, centrifugal force can balance gravity and prevent further inward radial collapse — the parcel will settle at a radius of order

$$R_{\mathrm{stall}} \sim \frac{j^2}{GM(< R)}.$$

This is the local condition is the same as the stall eradius we obtained in Part A.

Condition for rotational support

One can also express a rotational support condition in terms of energies. Let the rotational kinetic energy of the whole cloud be T_{rot}. The magnitude of gravitational energy is

$$|W| \sim \alpha\,\frac{GM^2}{R},$$

where α is a structure factor (e.g. $\alpha = 3/5$ for a uniform sphere, $\alpha \approx 1$ for more centrally concentrated distributions). The moment of inertia I can be written $I = kMR^2$ where k depends on mass distribution (for a uniform sphere $k = 2/5$).

Writing rotational KE in terms of specific angular momentum j. For a mass element distribution with typical j, the global rotational KE is

$$T_{\mathrm{rot}} \sim \frac{1}{2}I\Omega^2 = \frac{1}{2}kMR^2\Omega^2.$$

Since $j \sim R^2\Omega$ (for a parcel near R), $\Omega = j/R^2$. Substitute:

$$T_{\mathrm{rot}} \sim \frac{1}{2}kMR^2\left(\frac{j}{R^2}\right)^2 = \frac{1}{2}kM\frac{j^2}{R^2}.$$

Compare to $|W| \approx \alpha GM^2/R$. If rotation supplies enough kinetic energy to balance gravity in the virial sense, we would require (for full rotational support)

$$T_{\mathrm{rot}} \gtrsim \frac{|W|}{2} \quad \text{(since } T = -W/2 \text{ for equilibrium).}$$

Plugging in:

$$\frac{1}{2}kM\frac{j^2}{R^2} \gtrsim \frac{1}{2}\alpha\frac{GM^2}{R} \quad \Rightarrow \quad k\frac{j^2}{R^2} \gtrsim \alpha\frac{GM}{R}.$$

Rearrange:

$$j^2 \gtrsim \frac{\alpha}{k}\,GM\,R.$$

This is the global energy (virial) criterion. It has the same functional dependence $j^2 \propto GMR$ as the local force condition, but includes geometry factors α and k. For common simple models:

Uniform sphere: $\alpha = 3/5$, $k = 2/5 \Rightarrow \alpha/k = 3/2$. So the energy threshold becomes

$$j^2 \gtrsim \frac{3}{2}GMR$$

This is somewhat stricter than the purely local force condition $j^2 \gtrsim GMR$.

For a centrally concentrated mass distribution this factor changes, but the scaling remains.

Putting it all together

When the cloud collapses, it releases gravitational energy. About half of that energy (from the virial theorem) goes into motion — some into falling inward, some into faster rotation, and some into random or turbulent motions.

Angular momentum conservation ensures tangential kinetic energy grows faster with shrinking radius ($\propto R^{-2}$) than radial kinetic energy ($\propto R^{-1}$). Therefore rotation eventually dominates near small radii and halts radial collapse while axial collapse continues $\rightarrow$ flattening to a disk.

Two complementary criteria for rotational support:

- Local force balance: $j^2 \gtrsim GM(<R)\,R$.
- Global/virial (energy) balance: $j^2 \gtrsim \frac{\alpha}{k} GMR$.

The numerical prefactor depends on mass distribution; both show the same scaling $j^2 \propto GMR$.

PART C - FORCE BALANCE ANALYSIS

Force balance for a circular orbit

Consider a test mass m moving in a circular orbit of radius r around a central mass M. Two forces act in the radial direction:

Gravitational force (inward):

$$F_{\text{grav}} = \frac{GMm}{r^2}.$$

Centrifugal (inertial) force required for circular motion (outward in rotating-frame, or equivalently centripetal requirement for the inward force balancing the required acceleration):

$$F_{\text{cent}} = \frac{mv^2}{r},$$

where v is the tangential speed.

As shown in the figure below for a stable circular orbit these balance (net radial acceleration zero):

$$\frac{GMm}{r^2} = \frac{mv^2}{r}.$$

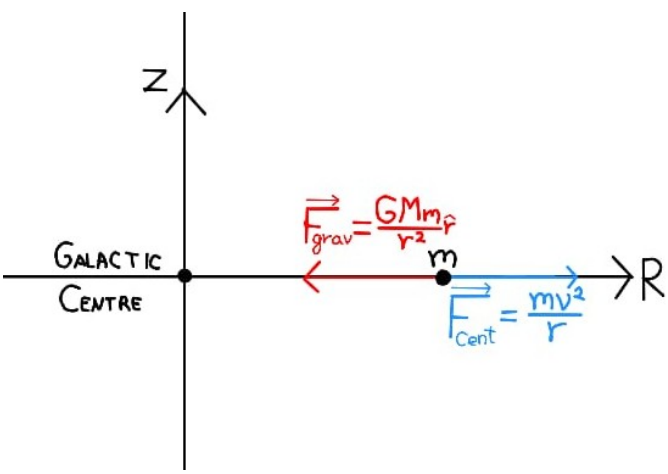

Fig. 10: Free-body force-balance sketch for a mass element m at radius r:

Cancel m and multiply both sides by r:

$$\frac{GM}{r} = v^2 \quad \Rightarrow \quad v = \sqrt{\frac{GM}{r}}.$$

This is the standard circular (Keplerian) orbital speed relation.

Expressing the balance in other forms

Angular speed

Define angular speed $\Omega = v/r$. From $v^2 = GM/r$,

$$\Omega^2 = \frac{v^2}{r^2} = \frac{GM}{r^3} \quad \Rightarrow \quad \Omega = \sqrt{\frac{GM}{r^3}}.$$

The angular speed required for centrifugal support at radius r equals the Keplerian (orbital) frequency $\sqrt{GM/r^3}$.

Angular momentum

Specific angular momentum about the axis is $j = rv = r^2\Omega$. Use $v = \sqrt{GM/r}$ to get the specific j of a circular orbit at radius r:

$$j_{\text{circ}}(r) = r\sqrt{\frac{GM}{r}} = \sqrt{GMr}.$$

Rewriting the local centrifugal-vs-gravity stall condition $j^2 \gtrsim GMr$ (from Parts A/B) recovers the same scaling: when a parcel's conserved j equals $\sqrt{GMr}$, circular/centrifugal support at radius r is possible.

Critical rotation rate for disk formation

Local (shell-by-shell) condition. A parcel with specific angular momentum j will stall at radius r_{stall} where its angular momentum equals the circular value:

$$j = j_{\text{circ}}(r_{\text{stall}}) = \sqrt{GM(<r_{\text{stall}})\,r_{\text{stall}}},$$

so

$$r_{\text{stall}} \sim \frac{j^2}{GM}.$$

If r_{stall} is much smaller than the initial cloud radius, material piles up there and a disk forms. Thus disk formation requires nonzero j so r_{stall} is not zero, the larger the initial j, the larger the radius where rotation halts infall.

Critical (breakup) angular speed at surface. If a self-gravitating object of mass M and characteristic radius R were to rotate as a rigid body, the angular speed at which centrifugal force at the equator equals gravity is

$$\Omega_{\text{crit}} = \sqrt{\frac{GM}{R^3}}.$$

(That is the same form as the Keplerian orbital frequency at the surface.) If the object rotated at or above Ω_{crit}, material at the equator would be centrifugally unbound (break-up). For disk formation you do not need the whole cloud to reach Ω_{crit}; you only need shells/parts to conserve enough angular momentum so that local centrifugal support stops inward motion at some smaller radius.

Global/energy (virial) estimate. From Part B, the global virial/energy condition for rotation to be dynamically important was

$$j^2 \gtrsim C\,GMR,$$

where $C = \alpha/k$ is an order-unity structural constant (depends on density profile). This is the same scaling in different clothes: whether looked at as force balance or energy balance, the threshold for rotation to halt collapse scales like $j^2 \sim GMR$.

PART D - NUMERICAL APPLICATION

Given

$$M = 10^{11}\,M_\odot, \quad R = 10\ \text{kpc}.$$

Constants used:

$$G = 6.67430 \times 10^{-11}\ \text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2},$$
$$M_\odot = 1.98847 \times 10^{30}\ \text{kg},$$
$$1\ \text{kpc} = 10^3\ \text{pc} = 3.08567758 \times 10^{19}\ \text{m}.$$

Circular (Keplerian) speed at radius R

Force-balance

$$\frac{GMm}{r^2} = \frac{mv^2}{r} \quad \Rightarrow \quad v^2 = \frac{GM}{r}.$$

$$v_c = \sqrt{\frac{GM}{R}}.$$

Numeric:

$$M = 10^{11} \times 1.98847 \times 10^{30} = 1.98847 \times 10^{41} \text{ kg},$$

$$\begin{aligned} R &= 10 \times 10^3 \text{ pc} \\ &= 10 \times 10^3 \times 3.08567758 \times 10^{16} \text{ m} \\ &= 3.08567758 \times 10^{20} \text{ m}, \end{aligned}$$

$$\begin{aligned} v_c &= \sqrt{\frac{6.67430 \times 10^{-11} \times 1.98847 \times 10^{41}}{3.08567758 \times 10^{20}}} \\ &\approx 2.0739 \times 10^5 \text{ m/s} \\ &= 207.39 \text{ km/s}. \end{aligned}$$

Angular frequency and orbital period

$$\Omega = \frac{v_c}{R}.$$

$$\Omega \approx 6.721 \times 10^{-16} \text{ s}^{-1}.$$

Orbital period

$$T_{\text{orb}} = \frac{2\pi}{\Omega} :$$

$$T_{\text{orb}} \approx 2.96 \times 10^8 \text{ years } (\approx 296 \text{ Myr}).$$

Gravitational binding energy (uniform sphere)

(Using $|W| = \frac{3}{5}\frac{GM^2}{R}$)

$$|W| \approx 5.13 \times 10^{51} \text{ J}, \quad T_{\text{vir}} \approx \frac{|W|}{2} \approx 2.57 \times 10^{51} \text{ J}.$$

Critical (Keplerian) angular speed for surface

$$\Omega_{\text{crit}} = \sqrt{\frac{GM}{R^3}} \approx 6.721 \times 10^{-16} \text{ s}^{-1},$$

(same numeric value as Ω above because we evaluated at the surface radius).

Specific angular momentum for a circular orbit at R

$$j_{\text{circ}} = \sqrt{GMR} \approx 6.399 \times 10^{25} \text{ m}^2/\text{s}.$$

we quote this usually in units $\text{kpc} \cdot \text{km/s}$:

$$j_{\text{circ}} \approx 2074 \text{ kpc km/s}.$$

MODULE-2

INTRODUCTION

Classifying galaxies based on their shapes — mainly into elliptical and spiral types — is very important in astronomy. It helps us understand how galaxies form and change over time. But big space surveys give us a lot of data without clear labels for each galaxy's type. This makes it hard to use regular supervised machine learning, which needs labeled examples.

In this module, we implemented a hybrid unsupervised + semi-supervised pipeline designed to:

- Learn galaxy representations directly from available features without using labels.
- Generate high-confidence pseudo-labels using clustering.
- Train supervised classifiers on these confident samples to improve accuracy.
- Incorporate domain knowledge to correctly map clusters to physical galaxy types.

The full implementation is available in our Kaggle notebook[11].

DATA PREPARATION AND FEATURE ENGINEERING

Splitting Data by Sky Location (skymap_id)

Each galaxy has a number called skymap_id which tells its position in the sky. Because different parts of the sky can have different conditions like brightness, stellar density, and atmospheric effects, the data from one part may look different from another. Mixing such regions can create domain shift, harming both clustering and classification performance.

So, we split the data into two groups:

- High skymap group: Galaxies with skymap_id greater than 130 million.
- Low skymap group: Galaxies with skymap_id less than or equal to 130 million.

This helped us train separate models that better understand the unique features of galaxies in each sky region.

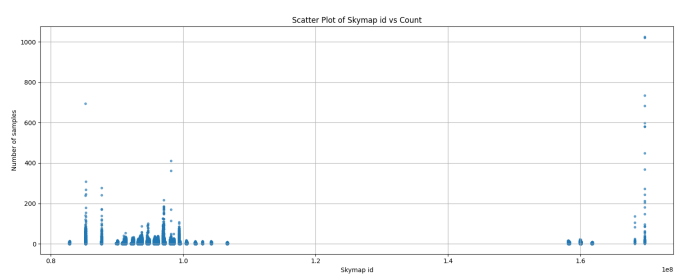

Fig. 11: Scatter Plot of Skymap id vs Count

Types of Features Used

We used two kinds of features for each galaxy:

(a) Simple Numerical Features: These are single numbers that describe a galaxy, like:

- Position in the sky (right ascension and declination)
- Size measurements (average size and area)
- Brightness values (average brightness and contrast)
- Color information (average color and color spread)
- Shape summaries and IDs

These numbers directly tell us about the galaxy's basic properties and can be used as input for machine learning models.

(b) Grouped List-Type Features:: Some features are stored as lists of numbers instead of single values. These lists capture information from different parts or angles of the galaxy, like:

- How bright the galaxy is at different radii (distance from center)
- Shape measurements taken at multiple angles
- Color values in different bins

Since these features have multiple numbers per galaxy, we process them carefully to keep all information.

Handling Missing or Incorrect Data in List Features

- Implementation: Checked each grouped feature; replaced non-list or invalid entries with empty lists.
- Reason: Ensures the feature matrices are consistent in shape for all galaxies, preventing errors during concatenation and model input.

Combining List Features into Matrices

We stacked the lists from each feature vertically so that each galaxy's data formed a row of numbers. Then, we joined all grouped features horizontally to make one big feature matrix.

For example, if we had 9 grouped features each with 3 numbers, we created a matrix with 27 numbers per galaxy. This way, all detailed multi-part features became one combined vector for each galaxy.

Final Feature Vector Creation

We combined the simple numerical features and the big grouped feature matrix side-by-side. This gave us one final feature vector for every galaxy, which had 46 numbers in total.

We created separate final feature sets for the high and low sky groups.

Scaling Features for Better Model Training

Standardization:

$$x' = \frac{x - \mu}{\sigma}$$

where μ and σ are the mean and standard deviation from the training data only.

Since different features had very different ranges (for example, brightness vs position), we scaled all features to have mean 0 and standard deviation 1.

This is called standardization. We calculated the scaling parameters (mean and std) from the training data only, and then applied the same scaling to validation and test data. Scaling helps machine learning models train faster and perform better by treating all features equally.

Approach 1: Unsupervised Embeddings + Clustering + Supervised Learning on Confident Samples

Step 1: Learning Embeddings from Training Data

For each sky group, we built a small neural network that learned to compress the input features into a 10-number vector called an embedding.

The network was trained without labels by making it reconstruct some parts of the input from others, so it learned meaningful patterns in the data.

These embeddings are simpler but capture important galaxy characteristics.

Step 2: Clustering Embeddings with KMeans

We applied KMeans clustering (with 2 clusters) on the training embeddings to separate galaxies into two groups, hoping that these groups correspond to elliptical and spiral galaxies.

This gave us "pseudo-labels" — guesses for galaxy types without using true labels.

Step 3: Selecting Confident Samples from Validation Data

We also applied clustering on the validation data embeddings and calculated how far each sample was from its assigned cluster center. Samples closer to their cluster centers were considered more confident.

We picked the top 50% most confident samples from validation data to use in the next step.

Step 4: Training a Supervised Classifier on Confident Samples

Using the confident samples and their pseudo-labels, we trained a simple neural network classifier to better separate the two galaxy types.

The classifier was trained with one-hot encoded labels to predict cluster membership.

Step 5: Predicting Test Data

Finally, we used the trained classifier to predict galaxy types on the test data for both sky groups, and combined their results.

Approach 2: Clustering on Validation Embeddings with Confidence Threshold Tuning

Motivation

Instead of clustering on training data, here we did clustering directly on validation embeddings, and tried different confidence thresholds to find the best pseudo-labels.

Step 1: Clustering on Validation Data

We ran KMeans clustering (2 clusters) on validation embeddings for both sky groups. We measured each sample's distance to cluster centers.

Step 2: Choosing Best Clustering Parameters

We tried different distance thresholds (like top 50th percentile) to select confident samples. The clustering that produced the highest number of confident samples was selected as best.

Step 3: Selecting Confident Samples

We picked confident samples below the median distance threshold to use for supervised training.

Step 4: Training Supervised Classifiers

On these confident pseudo-labeled samples, we trained a small neural network classifier similar to Approach 1.

Step 5: Test Set Prediction

The trained classifiers predicted labels on the test data, and we combined predictions for final output.

Understanding Clusters and Assigning Galaxy Types

To know which cluster corresponded to elliptical or spiral galaxies, we analyzed domain-specific features:

- Sersic Index: Elliptical galaxies usually have higher Sersic values (more concentrated light).
- Axis Ratio: Ellipticals are more spherical, so higher axis ratio.
- Color Variation: Spirals have more color variation due to star formation.

We normalized these features, combined them into an "elliptical score," and found which cluster had higher scores.

That cluster was labeled Elliptical, and the other Spiral. This gave physical meaning to our clusters.

Key Observations and Learnings

- Dividing data by sky region helped build better models for each area.
- Neural network embeddings learned good galaxy representations from raw features.
- Clustering produced useful pseudo-labels despite no real labels available.
- Filtering confident samples reduced noise in training.
- Training supervised classifiers on confident data improved classification accuracy.
- Domain knowledge helped map clusters to real galaxy types.

Conclusion

We successfully created a hybrid system that:

- Handles large unlabeled galaxy data using unsupervised and supervised methods.
- Splits data by sky location to improve model focus.
- Learns good feature embeddings automatically.
- Uses clustering and confidence filtering to generate training labels.
- Trains classifiers that accurately predict galaxy morphology.

This method can be very useful for future large sky surveys where labeled data is limited.

MODULE-3

PURPOSE OF THE PROJECT

The primary goal of this project is to simulate the formation and evolution of galaxies by modeling the motion of particles (representing stars, star clusters, or stellar mass points) under mutual gravitational attraction.

By adjusting the initial angular momentum L of the system, we can observe distinct large-scale galactic structures emerging:

- **High angular momentum** $\rightarrow$ spiral galaxies
- **Low angular momentum** $\rightarrow$ elliptical galaxies
- **Intermediate or asymmetric momentum** $\rightarrow$ irregular galaxies

This project demonstrates the connection between rotation and morphology in galaxy formation.

HOW THE SIMULATION WORKS

Particles

The simulation uses $N = 100$ particles. Each particle i has:

- Mass m_i (randomized within a small range to simulate mass diversity)
- Position $\vec{r}_i = (x_i, y_i)$ near the galactic center
- Velocity $\vec{v}_i = (v_{x,i}, v_{y,i})$ based on initial rotation parameters

Forces

Gravitational force between two particles i and j is given by Newton's law of gravitation with softening ϵ to prevent singularities:

$$\vec{F}_{ij} = -G \frac{m_i m_j (\vec{r}_i - \vec{r}_j)}{\left[|\vec{r}_i - \vec{r}_j|^2 + \epsilon^2\right]^{3/2}} \tag{1}$$

where:

- G — gravitational constant (set to 1 for normalization)
- ϵ — softening length to smooth extreme forces when $\vec{r}_i \approx \vec{r}_j$

Motion Update

We use the Euler integration method to update particle positions and velocities:

1) **Acceleration:**

$$\vec{a}_i = \frac{\vec{F}_{\text{total},i}}{m_i} \tag{2}$$

2) **Velocity Update:**

$$\vec{v}_i(t + \Delta t) = \vec{v}_i(t) + \vec{a}_i(t) \cdot \Delta t \tag{3}$$

3) **Position Update:**

$$\vec{r}_i(t + \Delta t) = \vec{r}_i(t) + \vec{v}_i(t + \Delta t) \cdot \Delta t \tag{4}$$

Here, Δt is the simulation time step.

Graphics & Plotting

- **Pygame:** Handles real-time particle animation, showing positions of particles as they evolve.
- **Matplotlib:** Plots total energy E_{total} and angular momentum L over time for physical accuracy monitoring.

$$E_{\text{total}} = \sum_i \frac{1}{2} m_i v_i^2 + \sum_{i<j} -\frac{G m_i m_j}{\sqrt{|\vec{r}_i - \vec{r}_j|^2 + \epsilon^2}} \tag{5}$$

$$L = \sum_i m_i \left(\vec{r}_i \times \vec{v}_i\right)_z \tag{6}$$

GALAXY TYPES SIMULATED

High Angular Momentum — Spiral Galaxy

- **Initial condition:** High tangential velocity for particles
- **Outcome:** Particles remain spread out, forming rotating spiral arms due to conservation of angular momentum.

Low Angular Momentum — Elliptical Galaxy

- **Initial condition:** Low tangential velocity
- **Outcome:** Particles fall toward the center, forming a dense, ellipsoidal distribution with minimal rotation.

Medium Angular Momentum + Randomness — Irregular Galaxy

- **Initial condition:** Moderate rotation with random velocity perturbations
- **Outcome:** Symmetry is broken, producing chaotic, uneven structures without well-defined arms.

OBSERVATIONS

- High rotation preserves radial spread and leads to structured spiral arms.
- Low rotation allows gravitational collapse, resulting in compact elliptical formations.
- Random velocity noise disrupts regular patterns, producing irregular galaxies.

Energy & Angular Momentum Tracking

- If E_{total} is constant $\rightarrow$ simulation is stable
- Any drift in E or L signals numerical integration error (common in Euler method).

Conclusion

This simulation effectively demonstrates the relationship between angular momentum and galaxy morphology. It integrates:

- **Physics:** Newtonian gravity with softening
- **Numerical methods:** Euler integration
- **Visualization:** Real-time graphics with Pygame
- **Scientific validation:** Energy and momentum conservation checks

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